

Project Design Phase-II Technology Stack (Architecture & Stack)

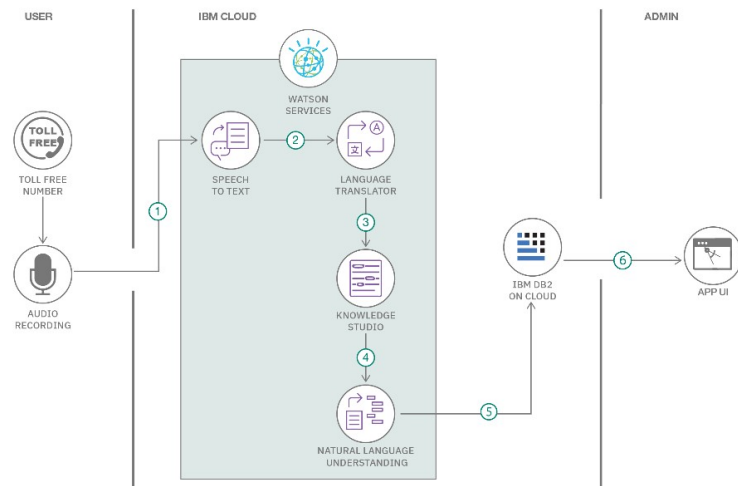
Date	31 January 3035
Team ID	SWTID-2026-3677
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Architecture Diagram: Auto Ticket Classification using Flow Designer

Technical Architecture: Auto Ticket Classification using Flow Designer

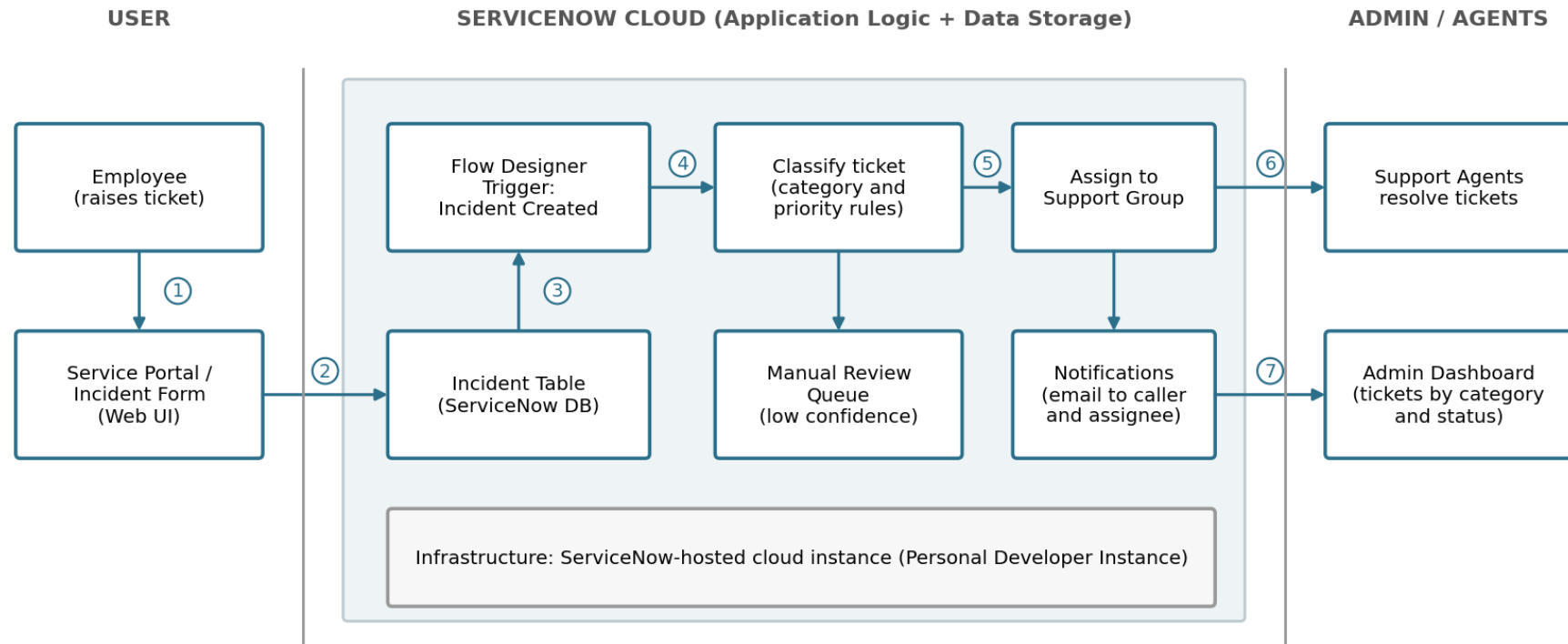


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employees raise incidents on the Service Portal; agents work tickets in the ServiceNow web UI	ServiceNow Service Portal, Now Experience UI (HTML, CSS, JavaScript)
2.	Application Logic-1	Flow Designer flow starts when an incident is created and reads its category and short description	ServiceNow Flow Designer (Trigger: Incident Created)
3.	Application Logic-2	Sets the priority from urgency keywords and assigns the ticket to the matching support group	Flow Designer actions and decision logic (Update Record)
4.	Application Logic-3	Notifies the caller and assignee and escalates tickets left unassigned	Flow Designer Send Email action, ServiceNow Notifications
5.	Database	Stores incident, assignment group and user records	ServiceNow platform tables (Incident, Group, User)
6.	Cloud Database	Database service on cloud hosting the ServiceNow instance	ServiceNow cloud-managed instance database
7.	File Storage	Ticket attachments	ServiceNow Attachment storage (platform)
8.	External API-1	Not applicable; email is sent through the ServiceNow email service	ServiceNow Email (SMTP)
9.	External API-2	Not applicable	Not applicable
10.	Machine Learning Model	Not applicable; classification is rule-based using Flow Designer	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on cloud Local Server Configuration: Not applicable Cloud Server Configuration: ServiceNow-hosted instance (Personal Developer Instance)	ServiceNow Cloud

Table-2: Application Characteristics:

S.No	Component	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks used	None; the solution is built on the ServiceNow low-code platform
2.	Security Implementations	Access control to tickets and flows using roles and ACLs; encrypted connection	ServiceNow roles and ACLs, HTTPS/TLS
3.	Scalable Architecture	Multi-tier cloud platform (presentation, application and database tiers)	ServiceNow cloud platform
4.	Availability	Application hosted and managed on the ServiceNow cloud	ServiceNow cloud hosting
5.	Performance	Flow runs automatically when a ticket is created, removing manual triage	Flow Designer, ServiceNow platform

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>